Prediction and Classification of virus families/sub-families, and Human-MERS/SARS-CoV/SARS-CoV-2 Protein-Protein Interactions Prediction using Machine Learning

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INTRODUCTION

Proteins usually need other proteins to be able to function or to be regulated. These reactions are called protein-protein interactions (PPI). It is important to know about these PPIs for drug target discovery because we can enhance or disrupt these interactions to relieve diseases. Predicting these interactions computationally is important to lower costs. We have developed a four-phase prediction and classification system for the virus families and sub-families, followed by the prediction criteria whether two protein sequences are a human-MERS/SARS1/SARS2 interaction with machine learning and neural networks, and the predictor will be used from a web server. The web server will work in four phases: it will predict whether the viral sequence is, indeed, a viral protein and, also, if it belongs to the viral class ssRNA(-), where Coronaviridae belongs; it will predict whether the protein comes from a coronavirus; it will predict whether the protein comes from MERS or SARS; and it will predict whether both proteins interact or not.

OBJECTIVES

1. Predict if a virus sequence is, indeed, a virus sequence, and if it is an ssRNA(-) class virus.
2. Predict if a virus sequence comes from a coronavirus.
3. Predict if a virus sequence comes from MERS or SARS.
4. Predict if a pair of MERS-SARS1-SARS2 and human proteins interact.

METHODS

- For phase 1 we gathered data from NCBI, VIPR and genome.jp.
- For phases 2 and 3 we gathered data from NCBI.
- For phase 4 we gathered experimentally tested interactions from multiple papers, Uniprot, IntAct, and BioGRID.
- For phase 4, we built the negative dataset crossing SARS1/SARS2/MERS proteins against human proteins, ranking the interactions by dissimilarity score between them and picking the interactions with lowest score.
- Sequences with less than 31 known amino acids were removed.
- For phase 3, we removed sequences with more than two or more consecutive Xs.
- For phases 1 to 3, we used CD-HIT to remove sequences with high similarity.
- We used iFeature for encoding protein sequences.
- We built neural networks using Keras.

DATA

Phase 1

Class	Total	Filtered
6 virus classes	3748182	711396
ssRNA(-)	7623595	106315
Non-virus	67530	66149

Phase 2

Class	Total	Filtered
Coronaviridae	1429752	385179
Negative	143948	49804

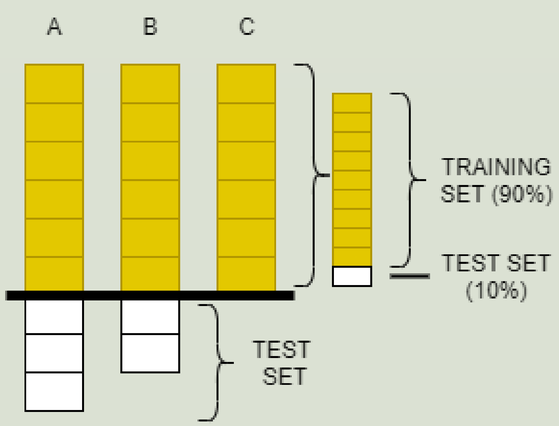
Phase 3

Class	Total	Filtered
SARS2	2972910	140516
SARS1	2977492	141435
MERS	8848	1694
Others	67530	2305

Phase 4

Class	Positive	Negative
SARS2	14001	20000
SARS1	1337	3000
MERS	318	1000

TRAINING AND TEST SETS



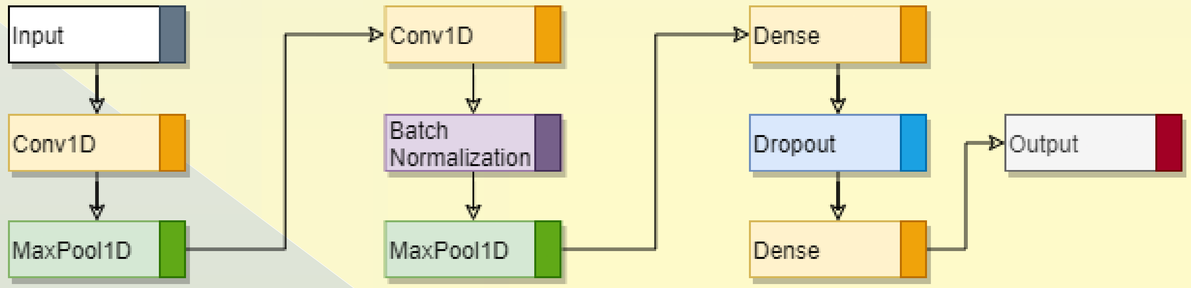
RESULTS

All tests were carried out with di-peptide composition (DPC) encoding.

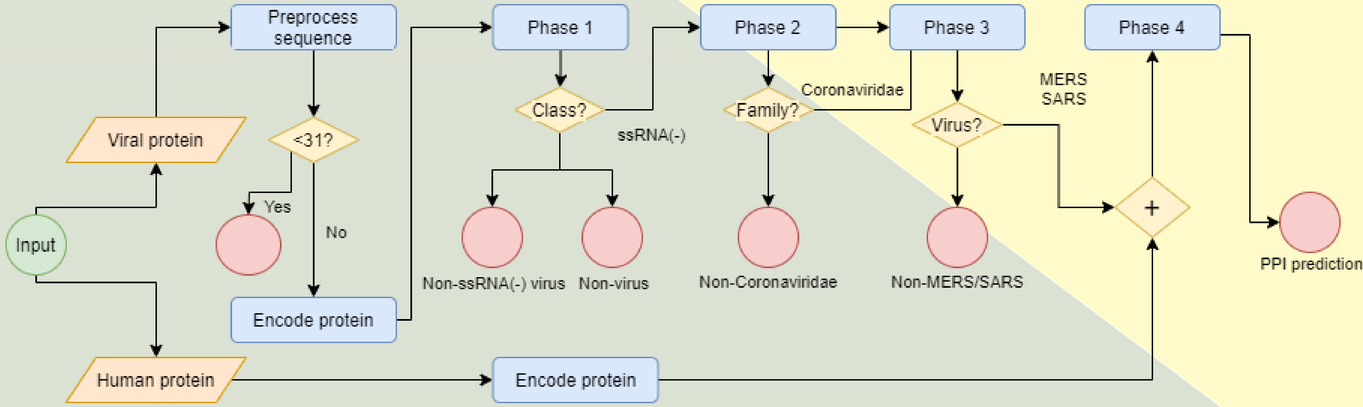
RF	MCC	ACC	F1
Phase 1	0.209	0.4874	0.5943
Phase 2	0.9921	0.9962	0.9962
Phase 3	0.9956	0.9981	0.997
Phase 4	0.8363	0.9317	0.9295

CNN	MCC	ACC	F1
Phase 1	0.7498	0.885	0.8991
Phase 2	0.9946	0,9981	0,9983
Phase 3	0.998	0.9992	0.9985
Phase 4	0.8469	0.9376	0.9366

CONVOLUTIONAL NEURAL NETWORK ARCHITECTURE



PREDICTION FLOW CHART



FUTURE RESEARCH

- Test with more encodings (i. e. AAC, QSOrder, NMBroto, PSSM) and combinations between them (i. e. DPC+NMBroto, PAAC+QSOrder).
- Improve network performance
- Develop the web server
- Add more virus-human interactions besides SARS1, SARS2 and MERS

CONTACT US

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